Notes: Crawford, Lee, Whinston, and Yurukoglu (Econometrica, 2018)

The Welfare Effects of Vertical Integration in Multichannel Television Markets

Contents

1 Big picture **3**

2 Data and measurement (institutional context and key objects) **4**

2.1 Industry structure and why this market is informative . . . 4
2.2 Regional sports networks (RSNs) and ownership variation . . . 4
2.3 Regulatory policy: Program Access Rules and the terrestrial loophole . . . 5
2.4 Downstream prices, quantities, channel lineups, and market construction . . . 5
2.5 Relevant RSNs, blackouts, and distance . . . 5
2.6 Viewership data . . . 6
2.7 Affiliate fees, advertising, and margins . . . 6
2.8 Descriptive facts . . . 6

3 Models and empirical specifications (all equations + interpretation) **7**

3.1 Economic mechanism and hypotheses . . . 7
3.2 Stage 3: household viewership . . . 7
3.3 Stage 2: household distributor choice . . . 8
3.4 Stage 1: distributor pricing and carriage . . . 8
3.5 Stage 1: bargaining, internalization, and rival foreclosure . . . 8
3.6 Estimation moments and instruments . . . 9
3.7 Counterfactual design . . . 10

4 Identification and empirical design (what makes the estimates credible) **10**

4.1 What identifies viewership and demand . . . 10
4.2 What identifies sports versus non-sports utility decay . . . 10
4.3 What identifies internalization and integrated carriage effects . . . 11
4.4 What identifies rival-foreclosure incentives . . . 11
4.5 How PARs enter the empirical design . . . 11
4.6 Why reverse causality and simple omitted-variables stories are less compelling . . . 12
4.7 What the paper can and cannot distinguish . . . 12
4.8 Relation to the literature . . . 12

5 Tables and figures **12**

5.1 Figure 1: RSN ownership . . . 12
5.2 Table A.I, A.II, and A.III: sample coverage, prices, ratings, and fees . . . 13
5.3 Table I and Figure 4: the raw integrated-carriage pattern . . . 13
5.4 Table II and Figure 5: key parameter estimates and willingness to pay . . . 15
5.5 Table III and Figure 3: demand elasticities, margins, and ESPN diversion . . . 16
5.6 Table IV: individual RSN counterfactuals . . . 17
5.7 Table V: average welfare effects across all RSNs . . . 17
5.8 Table VI and Table VII: when foreclosure hurts the most . . . 18

6 Short summary **18**

7 Conclusion **19**

7.1 Core conclusion (what the paper establishes) . . . 19
7.2 Economic intuition . . . 19
7.3 Takeaway . . . 19

1 Big picture

- **Question.** When cable or satellite distributors vertically integrate with regional sports networks (RSNs), does welfare rise because double marginalization falls and carriage expands, or does welfare fall because integrated firms foreclose rivals or raise their costs?
- **Setting.** The paper studies U.S. multichannel television markets from 2000 to 2010, with particular attention to high-value regional sports programming and the ownership of RSNs by cable or satellite distributors.
- **Why this setting is useful.** RSNs are valuable upstream inputs, ownership varies across regions and over time, and regulation created sharp variation in foreclosure possibilities through the Program Access Rules (PARs) and the terrestrial loophole.
- **Core challenge.** Higher carriage of integrated channels could reflect efficiency, while lower carriage of rival channels could reflect either efficient substitution or anticompetitive foreclosure. A convincing design must separately quantify pricing, carriage, bargaining, and welfare effects.
- **Main conceptual contribution.** The paper estimates a structural model that jointly captures household viewership, bundle choice, distributor pricing and carriage, and affiliate-fee bargaining. Crucially, it estimates rather than imposes how much integrated firms internalize each others' profits.
- **Main mechanisms.**
  - *Efficiency side:* integration can reduce double marginalization and increase carriage of the integrated RSN.
  - *Foreclosure side:* absent effective access rules, an integrated RSN may deny supply to rivals or charge them higher affiliate fees.
  - *Organizational side:* integrated firms may only partially internalize upstream–downstream profits, so vertical integration need not imply full joint-profit maximization.
- **Main empirical design.** The authors estimate the model using channel ratings, bundle market shares, prices, affiliate fees, carriage, and ownership patterns. They then simulate 2007 counterfactuals for 26 RSNs under three scenarios: no vertical integration, integration with PARs, and integration without PARs.
- **Main baseline result.** With PARs effectively enforced, integration of a single RSN lowers average cable prices by about **1.2%** (roughly \$0.67 per subscriber per month), raises cable carriage of the RSN by about **9.4%**, and increases total welfare by about **\$0.43 per household per month**.
- **Key internalization result.** Integrated firms internalize a substantial but incomplete share of partner profits: the estimate is $\mu \approx 0.79$. That is economically large, but far from the full internalization implied by simple unitary-firm models.
- **Key foreclosure result.** At the estimated lower bound for rival-foreclosure incentives, removing PARs leads to complete exclusion in **4 out of 26** RSNs; for the other 22 cases, integrated RSNs still supply rivals but raise their affiliate fees by about **18%**.

- **Net welfare result.** Even without PARs, average consumer and total welfare still rise relative to non-integration by about **1.3%** and **0.6%**, but rival satellite distributors lose market share and profits. The average masks important heterogeneity: gains are essentially absent in markets where exclusion occurs.
- **Economic takeaway.** The paper argues that the key policy question is not whether vertical integration is always bad, but whether regulation prevents integrated content owners from denying or overpricing access to rivals. Integration can be welfare enhancing on average, yet still harm rivals and create serious local losses in foreclosure-prone markets.

2 Data and measurement (institutional context and key objects)

2.1 Industry structure and why this market is informative

- During the sample period, most U.S. households could buy multichannel television from one local cable operator or from one of two nationwide satellite distributors, DirecTV and Dish.
- Distributors typically sold three bundles: limited basic, expanded basic, and digital. The analysis focuses on the expanded basic bundle, the most common bundle and the one that usually contains the channels of interest.
- Downstream distributors pay upstream channels monthly per-subscriber *affiliate fees*. These fees are central to the paper because integration changes how those fees matter for downstream pricing incentives.

Interpretation. The setting is a vertical market with differentiated downstream bundles and negotiated input prices. That makes it a natural laboratory for studying vertical integration.

2.2 Regional sports networks (RSNs) and ownership variation

- RSNs carry geographically local sports content, such as NESN in New England or MSG in New York.
- RSNs command high affiliate fees. The paper notes that in 2010 CSN Philadelphia averaged about $2.85 per subscriber per month, versus around $1 for channels like Fox News, TNT, and USA, and over $4 for ESPN.
- Among the RSNs active in 2007, downstream distributors held ownership stakes in 16. Cable owners included Comcast, Cablevision, Cox, and Time Warner, while DirecTV had indirect stakes through News Corporation and Liberty Media.

Interpretation. RSNs are important because they are both valuable and geographically localized. That combination creates variation in ownership, bargaining, and rivalry that a national channel setting would not provide.

2.3 Regulatory policy: Program Access Rules and the terrestrial loophole

- During the sample period, vertically integrated firms were subject to **Program Access Rules (PARs)** that required satellite-delivered integrated content to be supplied to rivals on nondiscriminatory terms, subject to arbitration if necessary.
- The rules did *not* cover content delivered terrestrially. This exemption, the **terrestrial loophole**, allowed some integrated cable RSNs to avoid mandatory supply to satellite rivals.
- In 2007, the two classic loophole cases were **CSN Philadelphia** and **4SD in San Diego**; in both cases the channel was not supplied to satellite distributors.
- The PARs later lapsed in 2012 and were replaced by a case-by-case "unfair acts" approach, but the paper's estimation period is 2000–2010.

Interpretation. The loophole markets are crucial because observed exclusion there provides the variation used to recover lower bounds on rival-foreclosure incentives.

2.4 Downstream prices, quantities, channel lineups, and market construction

- The foundational source is Nielsen FOCUS, which records cable-system lineups, homes passed, subscribers, ownership, and zip codes served.
- The paper uses years 2000–2010 and restricts attention to system-years without direct wireline cable competition.
- A **market** is defined as the set of zip codes served by a single cable system and, by construction, both satellite providers.
- Cable market shares combine FOCUS subscriber counts with MRI/Simmons survey data. Satellite market shares also come from MRI/Simmons.
- Satellite channel offerings and prices are collected from the Internet Archive.

Interpretation. The model is estimated at the local market level, but the market is built so that the cable distributor and the two satellite distributors are the relevant downstream competitors in each place.

2.5 Relevant RSNs, blackouts, and distance

An RSN is called *relevant* in a Nielsen DMA if at least 30% of the teams it carries are not blacked out there. The authors then assign that RSN as relevant to all markets in that DMA.

Two key RSN-specific demand shifters are:

$$b_{ict} = \text{fraction of teams on RSN } c \text{ that are blacked out for household } i, \tag{1}$$

$$d_{ic} = \text{distance from household } i \text{ to the teams' stadiums on RSN } c. \tag{2}$$

Interpretation. These variables are economically important because sports viewers value local teams more, and a channel is less useful if many of its games are blacked out.

2.6 Viewership data

- Household-level viewing comes from MRI (2000–2007) and Simmons (2008–2010), covering 96 to 99 national channels.
- Aggregate ratings come from Nielsen, which reports channel ratings for 44 to 56 large DMAs each year. Between 18 and 29 RSNs are covered depending on the year.
- The econometric analysis focuses on 38 national channels plus the RSNs.

Interpretation. Ratings and viewing frequencies identify how much consumers care about channels, while the share of households that ever watch a channel helps pin down the extensive margin of taste.

2.7 Affiliate fees, advertising, and margins

- SNL Kagan provides average affiliate fees and advertising revenues for national channels and RSNs.
- Average affiliate fees in the data are about **$0.30** per month for the 38 national channels and about **$1.64** for the RSNs.
- The paper also uses distributor margin moments from 2007 10-Ks: roughly 0.539 for Comcast, 0.396 for DirecTV, and 0.413 for Dish.

Interpretation. Affiliate fees matter for bargaining and pricing incentives, and the observed margins help identify demand elasticities and satellite taste heterogeneity.

2.8 Descriptive facts

- The estimation sample contains almost **7,000 market-years** across 2000–2010 and covers on average **39.7 million households** per year, more than 35% of U.S. households.
- Average cable price is about **$50.67** and average cable market share is about **0.628**.
- The average cable bundle includes about **67.0** cable networks, **1.6** RSNs, and **68.6** total channels.
- The average household subscribing to cable receives about **1.8 RSNs**, and around 80% of markets have one or two relevant RSNs available.

Interpretation. The data are rich enough to discipline both demand and supply. They combine local competition, local carriage, local prices, local relevance of sports content, and negotiated upstream fees.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper studies three linked supply-side decisions under vertical integration:

- bargaining over affiliate fees and whether supply occurs at all,
- downstream carriage of channels,
- downstream bundle pricing.

The main hypotheses are:

- integration can reduce *double marginalization* by making the downstream distributor internalize upstream RSN profits,
- integration can raise *carriage* of the owned RSN,
- absent PARs, integration can lead to *foreclosure* or *raising rivals' costs*.

Interpretation. All three margins move together in equilibrium, so the paper needs a model that solves them jointly rather than analyzing one reduced-form margin at a time.

3.2 Stage 3: household viewership

Household i that subscribes to MVPD f solves a time-allocation problem over channels:

$$\max_{w_{ift}} \; v_{ift}(w_{ift}) = \sum_{c \in B_{fmt} \cup \{0\}} \frac{\gamma_{ict}}{1-\nu_c} \left(w_{ifct}\right)^{1-\nu_c} \quad \text{s.t.} \quad \sum_{c \in B_{fmt} \cup \{0\}} w_{ifct} \leq T. \tag{3}$$

The taste parameter for channel c is

$$\gamma_{ict} = \begin{cases} \tilde{\gamma}_{ict}, & \text{with probability } \rho_c^0,\ \tilde{\gamma}_{ict} \sim \text{Exponential}(\rho_c^1), \\ 0, & \text{with probability } 1-\rho_c^0. \end{cases} \tag{4}$$

For RSNs, tastes are scaled by blackouts and distance:

$$\gamma_{ict}^{RSN} = \tilde{\gamma}_{ict} \exp(\gamma^b b_{ict} + \gamma^d d_{ic}). \tag{5}$$

Interpretation.

- ρ_c^0 captures how many households care at all about a channel.
- ρ_c^1 captures how much they care conditional on caring.
- Sports and non-sports channels are allowed to have different decay parameters, ν^S and ν^{NS}, because sports channels earn unusually high affiliate fees for a given rating.

3.3 Stage 2: household distributor choice

Conditional on the optimal viewing utility from a bundle, household utility from distributor f is

$$u_{ifmt} = \beta_v v^*_{ifmt} + \beta_x x_{fmt} + \beta^{sat}_{if} + \alpha p_{fmt} + \xi_{fmt} + \varepsilon_{ifmt}. \tag{6}$$

The resulting distributor choice probabilities are logit-style functions of price, bundle quality, and satellite taste heterogeneity.

Interpretation.

- v^*_{ifmt} is the value of the channels in the bundle.
- α is the price coefficient.
- β^{sat}_{if} lets some households especially like or dislike satellite service.
- ξ_{fmt} is an unobserved bundle-quality shock recovered by BLP inversion.

3.4 Stage 1: distributor pricing and carriage

Each downstream distributor chooses bundle prices and carriage to maximize profits. A simplified version of market-level profit is:

$$\Pi^M_{fmt} = D_{fmt}\left(p^{pre-tax}_{fmt} - mc_{fmt}\right) + \mu \times \big(\text{profits of integrated upstream channels affected by } f\big), \tag{7}$$

where marginal cost includes affiliate fees and non-channel costs,

$$mc_{fmt} = \sum_{c \in B_{fmt}} \tau_{fct} + \kappa_{fmt}. \tag{8}$$

Cable carriage satisfies

$$B_{fmt} = \arg\max_{B_f \subseteq A_{ft}} \Pi^M_{fmt}(\{B_f, B_{-fmt}\}, p_{mt}, \tau_t; \mu), \tag{9}$$

where A_{ft} is the set of channels with which the distributor has an agreement.

Interpretation.

- If $\mu = 0$, a distributor ignores upstream partner profits.
- If $\mu > 0$, an integrated distributor partly internalizes how lower prices or broader carriage help its owned RSN earn affiliate fees and ad revenue.

3.5 Stage 1: bargaining, internalization, and rival foreclosure

Affiliate fees are determined by Nash-in-Nash bargaining. For any distributor–channel pair,

$$\tau_{fct} = \arg\max \left(\mathrm{GFT}^M_{fct}\right)^{\zeta_{fct}} \left(\mathrm{GFT}^C_{fct}\right)^{1-\zeta_{fct}}, \tag{10}$$

where gains from trade depend on the downstream distributor's and upstream channel's surplus from reaching agreement.

The paper distinguishes:

- ζ^E: bargaining with *non-integrated* partners,
- ζ^I: bargaining within *integrated* pairs.

When an integrated RSN bargains with a non-integrated rival distributor, it may value the profit benefits from keeping viewers on its own downstream platform. This is governed by the rival-foreclosure parameter λ_R. In the paper's notation, the integrated channel's profit when bargaining with a rival includes a term proportional to

$$\mu\lambda_R \times \big(\text{lost downstream profit if the rival is supplied}\big). \tag{11}$$

Interpretation.

- If PARs bind, the model sets $\lambda_R = 0$: the integrated channel behaves as if it ignores foreclosure benefits when bargaining with rivals.
- If PARs do not bind, a larger λ_R makes exclusion or higher rival affiliate fees more likely.

3.6 Estimation moments and instruments

The parameters are estimated jointly by GMM using three blocks of moments.

(i) Viewership moments

- mean viewership for each national channel and RSN by year,
- the number of households with zero viewership for each channel-year.

(ii) Distributor-choice moments

- observed bundle market shares are matched using BLP inversion,
- price is instrumented with state satellite taxes,
- viewership utility is instrumented with the fraction of teams on relevant RSNs that are not blacked out.

(iii) Bargaining, markup, and carriage moments

- average affiliate fees for each RSN active in 2007 and four major national channels,
- implied margins for Comcast, DirecTV, and Dish,
- RSN carriage decisions by cable systems.

Interpretation. The estimation uses affiliate fees and margins not just as descriptive facts, but as equilibrium restrictions that discipline how valuable channels must be, how elastic demand must be, and how much integrated firms internalize each other.

3.7 Counterfactual design

For the 26 RSNs active in 2007, the paper computes equilibrium outcomes under three scenarios:

1. **No VI:** $\mu = 0$, $\lambda_R = 0$, so all firms behave as if non-integrated.
2. **VI with PARs:** integration is present, $\mu = \hat{\mu}$, but $\lambda_R = 0$.
3. **VI without PARs:** integration is present, $\mu = \hat{\mu}$, and λ_R is set to the larger estimated loophole lower bound.

Main simulations hold national satellite prices fixed, since the paper changes one RSN at a time. Later robustness checks allow satellite prices to adjust at the DMA level.

Interpretation. The comparison of (ii) versus (i) isolates efficiency gains from integration under effective access rules. The comparison of (iii) versus (ii) isolates the incremental harm from losing access rules.

4 Identification and empirical design (what makes the estimates credible)

4.1 What identifies viewership and demand

- Channel ratings and the share of households ever watching a channel identify the taste distribution parameters (ρ_c^0, ρ_c^1).
- Cross-market variation in blackout exposure and distance to stadiums identifies the RSN-specific utility shifters γ^b and γ^d.
- Variation in bundle prices and market shares identifies the price coefficient and the role of bundle viewership utility.
- Satellite taxes are used as a price instrument.

Interpretation. A key strength of the paper is that it does not rely only on market shares. It also uses actual viewing behavior and affiliate fees to discipline the demand side.

4.2 What identifies sports versus non-sports utility decay

The paper argues that sports channels earn much higher affiliate fees than non-sports channels with similar ratings. This pattern is visible in Figure 3 and is the main reason the model allows sports and non-sports channels to have different decay parameters.

Interpretation. If sports and non-sports had the same utility decay, the model would struggle to reconcile similar ratings with very different affiliate fees.

4.3 What identifies internalization and integrated carriage effects

The paper uses carriage behavior of integrated versus non-integrated systems. In Table I, the vertical-integration ownership-share coefficient is positive and significant, while distance to stadiums is strongly negative and the fraction of teams not blacked out is strongly positive.

A stylized version of the carriage evidence is:

$$\Pr(\text{carry RSN}) = a + b \cdot \text{VI ownership share} + c \cdot \text{blackout availability} + d \cdot \text{distance} + \varepsilon, \tag{12}$$

with $b > 0$, $c > 0$, and $d < 0$.

Interpretation. The crucial pattern is not simply that integrated systems carry more RSNs everywhere. Rather, integrated ownership matters especially in the intermediate-distance markets where carriage is most discretionary. That is the variation that pins down μ.

4.4 What identifies rival-foreclosure incentives

The paper estimates *lower bounds* for rival-foreclosure incentives using the two terrestrial-loophole markets, Philadelphia and San Diego, where integrated cable RSNs excluded satellite rivals.

The logic is straightforward:

- if exclusion is observed, then the gains from withholding must be at least as large as the gains from supplying the rival,
- therefore the minimum λ_R consistent with exclusion is identified as a lower bound.

The paper estimates lower bounds of about

$$\mu\lambda_R^{Phil} \approx 1.11, \qquad \mu\lambda_R^{SD} \approx 0.94. \tag{13}$$

Given $\mu \approx 0.79$, that implies λ_R is at least about 1.4 in Philadelphia and 1.3 in San Diego.

Interpretation. These are strong numbers. They imply that when bargaining with a rival distributor in loophole markets, integrated RSNs place substantial weight on the downstream benefits of foreclosure.

4.5 How PARs enter the empirical design

The model assumes:

$$\lambda_R = 0 \quad \text{when PARs are effectively enforced in non-loophole markets.} \tag{14}$$

Interpretation. This is a simplifying but central assumption. It means that the paper treats access rules as neutralizing foreclosure motives in bargaining, while leaving the efficiency side of integration intact.

4.6 Why reverse causality and simple omitted-variables stories are less compelling

- Price endogeneity is addressed with satellite-tax instruments.
- Viewership utility is instrumented with blackout variation.
- The higher carriage of integrated RSNs is studied conditional on distance and blackout differences, which absorb much of the raw heterogeneity in sports relevance across places.
- Foreclosure is anchored by observed total exclusion in specific loophole markets rather than by broad cross-sectional correlations.

Interpretation. The paper's credibility does not come from a single reduced-form shock. It comes from a system of mutually consistent moments: shares, prices, ratings, affiliate fees, carriage, and loophole exclusions all have to line up under one parameter vector.

4.7 What the paper can and cannot distinguish

- It studies foreclosure of downstream distributors by integrated channels, not the foreclosure of upstream rival channels by distributors.
- It does not allow vertical integration to alter content investment or distribution-service quality.
- It takes ownership structures as given and does not model why specific integrations occurred.
- It is a partial-equilibrium analysis of one-RSN changes at a time, with national satellite prices fixed in the main simulations.

Interpretation. The paper is strongest on pricing, carriage, bargaining, and access. It is not a general model of long-run industry evolution or of dynamic investment incentives.

4.8 Relation to the literature

Earlier reduced-form studies showed that integrated systems tend to carry their own content more often, but they could not separate efficiency from foreclosure. This paper's main contribution is to decompose those channels structurally and attach welfare numbers to them.

5 Tables and figures

5.1 Figure 1: RSN ownership

Read:

- The figure tracks ownership of the RSNs active in 2007 across 1998–2010.

- It shows substantial heterogeneity across regions and time, with both cable and satellite links to RSNs.
- In 2007, 16 of the active RSNs had some downstream ownership affiliation.

Economic message. Ownership variation is the raw material for the whole paper. Without cross-RSN and over-time variation in who owns which RSN, the paper could not estimate integration effects or simulate alternative ownership structures.

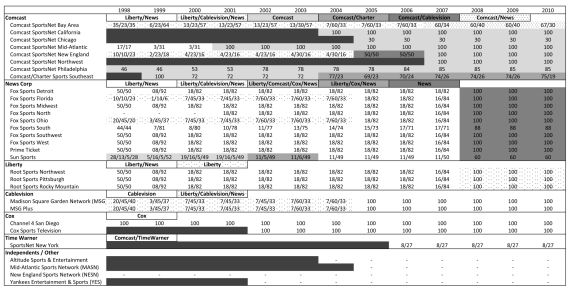


FIGURE 1.—RSN ownership. *Notes*: Reported are the vertical ownership stakes held by major distributors of cable and satellite television service for the Regional Sports Networks (RSNs) in our data that were active in 2007. Ownership data were collected by hand from company stock filings and industry sources. The ownership share for each distributor is reported and individual owners (or combinations of owners) are shaded according to the channel-specific legend located in the first line of each potential owner. Black bars correspond to a year in which the given RSN is not active (i.e., has not yet entered or has exited the market). Hyphens correspond to years of active operation for an RSN without a vertical ownership affiliation.

TABLE A.I

SAMPLE STATISTICS—PRICES, MARKET SHARES, AND CHANNELS[a]

		Unweighted				Weighted by HHs			
	# Obs	Mean	StdDev	Min	Max	Mean	StdDev	Min	Max
Total Markets	6,928	6,928							
Average Households (millions)	6,928					39.7			
Cable									
Year	6,928	2004	2.9	2000	2010	2004	2.8	2000	2010
Price	6,928	$50.67	$10.31	$8.67	$130.96	$52.20	$8.86	$8.67	$130.96
Market Share	6,928	0.628	0.162	0.001	0.965	0.639	0.135	0.001	0.965
Cable Networks	6,928	67.0	18.1	0	101	72.1	14.8	0	101
RSNs	6,928	1.6	0.8	0	5	1.8	0.9	0	5
Total Channels	6,928	68.6	18.4	0	103	73.8	15.0	0	103
DirecTV									
Year	6,928	2004	2.9	2000	2010	2004	2.8	2000	2010
Price	6,928	$52.76	$6.36	$46.05	$76.73	$52.66	$6.08	$46.05	$76.73
Market Share	6,928	0.090	0.060	0.002	0.499	0.091	0.063	0.002	0.499
Cable Networks	6,928	79.4	10.4	66	97	79.8	10.2	66	97
RSNs	6,928	1.9	1.0	0	7	1.9	1.0	0	7
Total Channels	6,928	84.3	10.9	69	107	84.7	10.7	69	107
Dish									
Year	6,928	2004	2.9	2000	2010	2004	2.8	2000	2010
Price	6,928	$53.43	$4.81	$44.28	$68.33	$53.47	$4.63	$44.28	$68.33
Market Share	6,928	0.062	0.054	0.000	0.472	0.058	0.052	0.000	0.472
Cable Networks	6,928	69.4	13.3	54	91	70.0	13.1	54	91
RSNs	6,928	1.8	0.9	0	6	1.7	0.8	0	6
Total Channels	6,928	73.7	13.9	56	99	74.4	13.6	56	99

[a]Reported are the average price, market share, and number of cable networks, Regional Sport Networks (RSNs), and total channels for the local cable operators as well as the two national satellite providers serving each of our markets. Markets are defined as the set of continuous zip codes within a cable system facing the same portfolio of competitors. We exclude (the relatively few) markets facing competition between cable operators. All the data cover the years 2000–2010. To be included, we required information on each of price, market share, and channels. Cable system subscriber and channel information is from the Nielsen FOCUS data set. Cable system price information is drawn from the Internet Archive, newspaper reports, and the TNS Bill Harvesting database. Satellite system channel and price information is drawn from the Internet Archive. Cable and satellite subscriber market shares are estimated from the MRI (2000–2007) and Simmons (2008–2010) household surveys and Nielsen FOCUS data set. See the text for more details.

5.2 Table A.I, A.II, and A.III: sample coverage, prices, ratings, and fees

Read:

- The estimation sample covers almost 7,000 market-years and about 39.7 million households per year.
- Cable average price is about $50.67 and average market share about 0.628.
- The two satellite distributors charge around $53 on average and have more elastic demand than cable.
- National channels have average affiliate fees around $0.30, versus about $1.64 for RSNs.

Economic message. RSNs are not small side products. They are expensive, valuable, and plausibly important enough to move downstream pricing and carriage incentives.

5.3 Table I and Figure 4: the raw integrated-carriage pattern

Read:

TABLE A.II

NATIONAL CABLE CHANNELS: AFFILIATE FEES AND VIEWERSHIP[a]

	Affiliate Fees					Viewership					
	Kagan					Nielsen Ratings		Combined MRI / Simmons			
	Years	Mean	StDev	Min	Max	Obs	Mean	Obs	Mean	SDev	Frac > 0
ABC Family Channel	11	$0.19	$0.02	$0.16	$0.22	747	0.418	277,535	0.344	1.149	0.176
AMC	11	$0.22	$0.02	$0.20	$0.25	747	0.491	277,535	0.351	1.183	0.156
Animal Planet	11	$0.07	$0.01	$0.06	$0.09	747	0.275	277,535	0.344	1.108	0.203
A&E	11	$0.21	$0.03	$0.16	$0.26	747	0.664	277,535	0.472	1.373	0.230
BET	11	$0.14	$0.02	$0.11	$0.17	747	0.382	277,535	0.184	1.017	0.070
Bravo	11	$0.15	$0.03	$0.11	$0.20	747	0.277	277,535	0.169	0.804	0.092
Cartoon Network	11	$0.14	$0.03	$0.08	$0.18	747	0.989	277,535	0.231	1.098	0.106
CMT	11	$0.06	$0.02	$0.01	$0.08	747	0.142	277,535	0.120	0.732	0.067
CNBC	11	$0.24	$0.04	$0.16	$0.30	747	0.217	277,535	0.313	1.185	0.170
CNN	11	$0.43	$0.05	$0.35	$0.52	747	0.550	277,535	0.701	1.744	0.319
Comedy Central	11	$0.11	$0.02	$0.08	$0.14	747	0.449	277,535	0.280	0.997	0.162
Discovery Channel	11	$0.27	$0.04	$0.22	$0.35	747	0.535	277,535	0.628	1.462	0.327
Disney Channel	11	$0.81	$0.06	$0.75	$0.91	747	1.171	277,535	0.246	1.074	0.116
E! Entertainment TV	11	$0.19	$0.02	$0.15	$0.21	747	0.315	277,535	0.201	0.788	0.137
ESPN	11	$2.81	$1.12	$1.14	$4.34	747	0.836	277,535	0.675	1.767	0.257
ESPN 2	11	$0.37	$0.14	$0.17	$0.58	747	0.262	277,535	0.334	1.220	0.151
ESPN Classic Sports	11	$0.14	$0.03	$0.10	$0.18	636	0.037	277,535	0.072	0.521	0.047
Food Network	11	$0.06	$0.03	$0.03	$0.14	747	0.411	277,535	0.396	1.364	0.175
Fox News Channel	11	$0.32	$0.18	$0.17	$0.70	747	0.785	277,535	0.697	1.961	0.267
FX	11	$0.34	$0.06	$0.27	$0.43	747	0.463	277,535	0.258	0.976	0.137
Golf Channel	11	$0.20	$0.05	$0.13	$0.26	580	0.065	277,535	0.084	0.633	0.041
Hallmark Channel	11	$0.04	$0.02	$0.01	$0.06	699	0.307	225,618	0.301	1.268	0.088
Headline News	—	—	—	—	—	747	0.214	277,535	0.278	0.983	0.173
HGTV	11	$0.08	$0.04	$0.03	$0.14	747	0.500	277,535	0.397	1.446	0.162
History Channel	11	$0.18	$0.04	$0.13	$0.23	747	0.531	277,535	0.531	1.462	0.251
Lifetime	11	$0.21	$0.06	$0.13	$0.29	747	0.679	277,535	0.554	1.650	0.199
MSNBC	11	$0.14	$0.02	$0.12	$0.17	747	0.343	277,535	0.330	1.181	0.182
MTV	11	$0.27	$0.05	$0.20	$0.35	747	0.568	277,535	0.235	0.983	0.127
Nickelodeon	11	$0.37	$0.05	$0.29	$0.47	747	1.555	277,535	0.200	0.991	0.096
SyFy	11	$0.17	$0.04	$0.12	$0.22	747	0.427	277,535	0.301	1.207	0.126
TBS	11	$0.37	$0.12	$0.19	$0.54	747	0.905	277,535	0.497	1.345	0.243
TLC	11	$0.16	$0.01	$0.14	$0.17	747	0.422	277,535	0.342	1.151	0.173
truTV	11	$0.09	$0.01	$0.08	$0.10	747	0.384	277,535	0.233	1.081	0.101
Turner Classic Movies	11	$0.22	$0.03	$0.16	$0.27	580	0.286	277,535	0.268	1.142	0.105
TNT	11	$0.83	$0.16	$0.55	$1.10	747	1.219	277,535	0.592	1.553	0.263
USA	11	$0.46	$0.07	$0.36	$0.57	747	1.081	277,535	0.503	1.442	0.230
VH1	11	$0.12	$0.02	$0.09	$0.16	747	0.336	277,535	0.151	0.717	0.101
Weather Channel	11	$0.10	$0.01	$0.08	$0.12	747	0.234	277,535	0.380	1.046	0.326

[a] Reported are average affiliate fees and viewership of the 38 non-RSN cable television networks included in our demand system. The averages are over the years 2000–2010 for SNL Kagan affiliate fees, over DMA-years for the Nielsen (DMA-level) viewership data, and over households and years for the MRI (2000–2007) and Simmons (2008–2010) household-level viewership data. Affiliate fees are the monthly per subscriber fees paid by cable and satellite distributors to television networks for the right to distribute the network's programming to subscribers. The Nielsen "rating" is the percentage of U.S. households watching a given program on a given channel at a given time. We average program-level ratings across programs within a channel-DMA-year, and report the across-DMA-year average here. MRI/Simmons viewership is reported as the average number of hours watching that channel in a typical week. It is converted to a Nielsen-equivalent "rating" by dividing by the number of hours in a week and rescaling it to lie between 0 and 100. The average fraction of households viewing a channel at all ("Frac > 0") is the average percentage of households that reported positive viewing of a channel in a typical week.

TABLE A.III

REGIONAL SPORTS NETWORKS: AVAILABILITY, AFFILIATE FEES, AND VIEWERSHIP[a]

	Kagan Availability		Kagan Affiliate Fees					Nielsen Viewing				
	Systems Served	HH Served	Years	Mean	StDev	Min	Max	# Obs	All HH	Has Cable	Has DTV	Has Dish
Comcast RSNs												
CSN Bay Area	137	4.7	11	$1.70	$0.53	$1.01	$2.52	720	0.41	0.48	0.45	0.33
CSN California	135	10.0	7	$0.91	$0.14	$0.75	$1.10	360	0.22	0.26	0.20	0.22
CSN Chicago	335	5.7	7	$2.02	$0.18	$1.90	$2.37	360	0.54	0.66	0.59	0.36
CSN Mid-Atlantic	194	6.2	11	$2.03	$0.74	$0.85	$3.10	1,440	0.13	0.17	0.09	0.03
CSN New England	222	4.8	11	$1.26	$0.32	$0.90	$1.89	1,080	0.27	0.29	0.30	0.17
CSN Philadelphia	102	4.6	11	$1.94	$0.61	$1.05	$2.85	360	0.91	1.15	0.06	0.05
News Corp RSNs												
Fox Sports Detroit	284	5.3	11	$1.75	$0.45	$1.05	$2.34	360	1.02	1.25	0.94	0.68
Fox Sports Florida	152	6.7	11	$1.34	$0.33	$0.90	$1.95	2,160	0.14	0.16	0.12	0.12
Fox Sports Midwest	695	7.4	11	$1.42	$0.44	$0.57	$2.01	1,800	0.31	0.40	0.31	0.26
Fox Sports North	620	4.5	11	$1.97	$0.60	$1.15	$2.88	720	0.79	1.01	1.04	0.70
Fox Sports Ohio	306	7.0	11	$1.61	$0.49	$0.75	$2.42	2,160	0.34	0.42	0.31	0.29
Fox Sports South	905	15.3	11	$1.40	$0.52	$0.52	$2.17	2,880	0.14	0.19	0.09	0.08
Fox Sports Southwest	924	12.7	11	$1.68	$0.50	$0.80	$2.43	4,320	0.16	0.21	0.17	0.13
Fox Sports West	167	9.2	11	$1.80	$0.44	$0.87	$2.35	1,080	0.16	0.21	0.12	0.07
Prime Ticket (New)	132	8.2	11	$1.52	$0.46	$0.60	$2.07	720	0.16	0.22	0.12	0.09
Sun Sports	234	8.3	11	$1.36	$0.54	$0.55	$2.27	1,440	0.18	0.00	0.11	0.09
Liberty RSNs												
Root Sports Northwest	281	5.4	11	$1.73	$0.52	$0.70	$2.54	256	0.23	0.30	0.19	0.15
Root Sports Pittsburgh	316	4.5	11	$1.81	$0.53	$1.05	$2.55	512	0.06	0.07	0.06	0.03
Root Sports Rocky Mountain	479	5.4	11	$1.58	$0.42	$0.75	$2.06	384	0.08	0.10	0.10	0.13
Cablevision RSNs												
Madison Square Garden Network (MSG)	219	9.9	11	$1.82	$0.30	$1.45	$2.44	1,080	0.23	0.07	0.24	0.17
MSG Plus	165	7.5	11	$1.24	$0.15	$1.01	$1.61	360	0.07	0.27	0.05	0.06
Cox RSNs												
Channel 4 San Diego	15	1.0	11	$0.87	$0.26	$0.53	$1.32	360	0.48	0.60	0.03	0.00
Time Warner RSNs												
SportsNet New York	314	20.1	5	$1.91	$0.18	$1.71	$2.20	1,080	0.13	0.15	0.13	0.09
Independent/Other RSNs												
Altitude Sports & Entertainment	130	2.8	7	$1.99	$0.29	$1.70	$2.47	360	0.24	0.30	0.21	0.22
Mid-Atlantic Sports Network (MASN)	109	5.2	6	$1.58	$0.12	$1.45	$1.77	1,800	0.11	0.13	0.08	0.11
New England Sports Network (NESN)	213	4.5	11	$1.99	$0.49	$1.30	$2.72	1,080	0.95	1.02	1.00	0.48

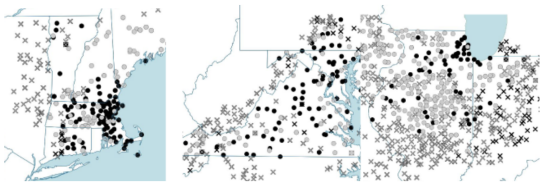
[a] Reported are the average availability, affiliate fees, and viewing of the 26 Regional Sports Networks (RSNs) active in 2007 that we focus on in our analysis. "Systems served" is the number of systems carrying the RSN, and "HH served" is the number of households to which the RSN is available. Affiliate fees are the monthly per subscriber fees paid by cable and satellite distributors to television networks for the right to distribute the network's programming to subscribers. Availability and affiliate fee information is provided by SNL Kagan. RSN viewership is provided by Nielsen. For availability and affiliate fees, the average is over the years 2000–2010, and for viewing the average is over DMA-years between 2007 and 2011 among the top 56 DMAs for which the RSN is deemed to be "relevant" (which occurs if at least 30 percent of the teams carried by that RSN are not "blacked out"; see text for further details). Nielsen viewing is the average "rating," the percentage of U.S. households watching a given program on a given channel at a given time, reported for all U.S. households as well as for those households that subscribe to cable or to the two national satellite providers, DirecTV (DTV) and Dish.

TABLE I

REGRESSION OF RSN CARRIAGE ON INTEGRATION STATUS, DISTANCE, AND BLACKOUT PERCENTAGE[a]

	(1)	(2)	(3)	(4)
VI Ownership Share	0.404***	0.435***	0.293***	0.171**
	(0.0674)	(0.0837)	(0.110)	(0.0852)
% Teams Not Blacked Out	0.412***	0.399***	0.429***	0.477***
	(0.0494)	(0.0586)	(0.109)	(0.107)
Avg. Distance to RSN's Stadiums (10^3 mi)	−0.559***	−0.630***	−0.838***	−0.795***
	(0.100)	(0.117)	(0.238)	(0.284)
Years	2000–2010	2007	2007	2007
Systems	All Systems	All Systems	Has P Q	Has P Q
Has Deal	No	No	No	Yes
Observations	154,121	12,246	1,132	1,052
R-squared	0.615	0.616	0.670	0.639

[a] Linear probability regression where the dependent variable is whether a cable system carries an RSN in a relevant market in 2007. Specifications differ by sample used, where "Has P Q" restricts attention to systems for which price and quantity data are available, and "Has Deal" restricts attention to system-RSN pairs where the MVPD has a deal with the RSN (i.e., carries the RSN on at least one other system). All specifications use DMA, RSN, and (when appropriate) year fixed effects. Inclusion of system demographic controls (race, population density, average income, household ownership) did not appreciably change point estimates. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Standard errors are reported in parentheses, and are clustered by DMA.



(a) CSN New England (b) CSN Mid-Atlantic (c) CSN Chicago

FIGURE 4.—Carriage by Comcast and non-integrated cable MVPDs of three Comcast-integrated RSNs across cable systems in 2007. Dots represent carriage by a system, X's represent no carriage. Black markers represent Comcast systems, grey markers represent non-Comcast cable systems.

- Table I regresses RSN carriage on integrated ownership share, blackout exposure, and distance.
- The vertical-integration coefficient is positive and statistically significant, ranging from about 0.17 to 0.44 across specifications.
- The fraction of teams not blacked out enters positively, around 0.40 to 0.48.
- Average distance to teams' stadiums enters negatively, roughly from -0.56 to -0.80 per 1,000 miles.
- Figure 4 shows the same logic visually: integrated Comcast systems are more likely to carry owned RSNs in moderate-distance markets than nonintegrated systems.

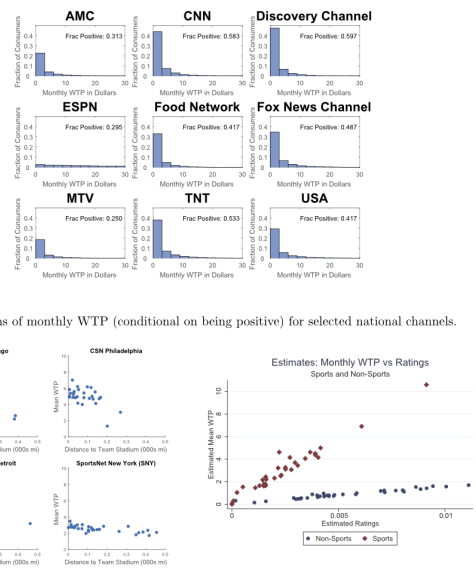
Economic message. This is the cleanest reduced-form fact behind the estimate of μ: integration raises carriage of the owned RSN beyond what a purely non-integrated profit calculation would suggest.

5.4 Table II and Figure 5: key parameter estimates and willingness to pay

TABLE II
ESTIMATES OF KEY PARAMETERS[a]

	Parameter	Description	Estimate	SE
Viewership Parameters $\boldsymbol{\theta}_1$	ν^{NS}	Viewership Decay, Non-sports	0.59	0.00
	ν^{S}	Viewership Decay, Sports	0.95	—
	γ^{v}	Fraction of Teams Blacked-out	−0.58	0.31
	γ^{d} (10^3 mi)	Distance	−0.93	0.27
Bundle Choice Parameters $\boldsymbol{\theta}_2$	α	Bundle Price	−1.00	0.44
	β^{v}	Bundle Viewership Utility	0.14	0.07
	$\rho^{\text{int}}_{\text{DirecTV}}(10^2)$	DirecTV Exponential Parameter	0.42	0.23
	$\rho^{\text{int}}_{\text{Dish}}(10^2)$	Dish Exponential Parameter	0.49	0.27
Pricing, Bargaining, Carriage and Foreclosure Parameters $\boldsymbol{\theta}_3, \lambda_R$	σ^2_{ω}	Variance of Carriage Shocks	0.00	0.00
	ζ^{E}	Bargaining, External	0.28	0.03
	ζ^{I}	Bargaining, Internal	0.37	0.06
	μ	Internalization	0.79	0.09
	$\mu \times \underline{\lambda}^{\text{Phil}}_R$	Internalization & Rival Foreclosure, Philadelphia	1.11	0.14
	$\mu \times \underline{\lambda}^{\text{SD}}_R$	Internalization & Rival Foreclosure, San Diego	0.94	0.11

[a]Selected key parameters from the first and second step estimation of the full model, where parameter ν^S is estimated separately via a grid search (see Appendix S.B.3 of the Supplemental Material). Additional viewership parameters contained in $\boldsymbol{\theta}_1$ are reported in Appendix Table A.IV; state-firm and year fixed effects in $\boldsymbol{\theta}_2$ are not reported. Asymptotic GMM standard errors are computed using numerical derivatives and 1500 bootstrap draws of markets and simulated households to estimate the variance-covariance matrix of the moments. Estimates and standard errors for $\mu \times \underline{\lambda}^{\text{Phil}}_R$ and $\mu \times \underline{\lambda}^{\text{SD}}_R$ are for the lower bound of these parameters.



(a) Histograms of monthly WTP (conditional on being positive) for selected national channels.

(b) Mean WTP in a market versus distance to RSN's teams' stadiums, for four RSNs.

(c) Estimated monthly WTP versus estimated ratings for sports and non-sports channels.

FIGURE 5.—Predicted willingness-to-pay (WTP) for channels (2007 values).

Read:

- The estimated price coefficient is about $\alpha = -1.00$, and the coefficient on bundle viewership utility is about $\beta_v = 0.14$.
- The estimated internalization parameter is $\mu = 0.79$.
- External and internal bargaining parameters are about $\zeta^E = 0.28$ and $\zeta^I = 0.37$.
- The lower bounds on $\mu\lambda_R$ are about 1.11 in Philadelphia and 0.94 in San Diego.

- Figure 5 shows that most national channels have mean WTP below $1 per month, while many RSNs have mean WTP above $2, and some much larger.

Economic message. The estimates imply that sports content is special, integrated firms only partially internalize joint profits, and observed loophole exclusion requires a strong foreclosure motive.

5.5 Table III and Figure 3: demand elasticities, margins, and ESPN diversion

TABLE III
ELASTICITIES AND MARGINS[a]

Elasticity of Row With Respect to Price of Column:	Cable	DirecTV	Dish
Cable	−1.69	0.30	0.19
DirecTV	2.16	−2.90	0.13
Dish	3.18	0.22	−4.15
Outside Option	5.52	0.26	0.16
Predicted Margins			
Mean Comcast Margin		0.66	
Mean DirecTV Margin		0.48	
Mean Dish Margin		0.45	
Logit Demand Price Coefficients			
OLS Logit Price Coefficient		−0.004** (0.002)	
IV Logit Price Coefficient		−0.080*** (0.025)	
Semi-Elasticity of Row With Respect to Removal of ESPN From column:	**Cable**	**DirecTV**	**Dish**
Cable	−18.90	3.86	2.36
DirecTV	54.82	−19.52	1.11
Dish	51.16	1.85	−19.67
Outside Option	17.27	0.22	0.14

[a]This table reports predicted mean price elasticities, predicted margins for Comcast and the two satellite distributors, the estimated price coefficient from a logit demand regression without (OLS) and with (IV) the use of price instruments (where standard errors clustered at market level), and semi-elasticities from dropping the national channel ESPN. For logit demand estimates, **, *** represent significance at the 5% and 1% levels.

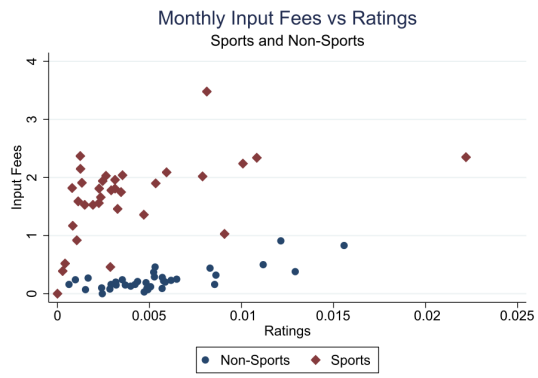


FIGURE 3.—Negotiated monthly affiliate fees and viewership ratings.

Read:

- Mean own-price elasticities are about −1.69 for cable, −2.90 for DirecTV, and −4.15 for Dish.
- Predicted margins are about 0.66 for Comcast, 0.48 for DirecTV, and 0.45 for Dish.
- In a logit demand regression, instrumenting for price changes the price coefficient from about −0.004 in OLS to about −0.080 in IV.
- If DirecTV lost ESPN, its market share would fall by about 19.5%; most of those households would switch to cable rather than to Dish or the outside option.
- Figure 3 shows that sports channels receive much higher affiliate fees than non-sports channels with similar ratings.

Economic message. Satellite demand is relatively elastic and cable is the main substitute. That is exactly the environment in which a cable-owned RSN can gain from worsening a satellite rival's offer.

5.6 Table IV: individual RSN counterfactuals

Read:

- **CSN Philadelphia** illustrates a loophole market. With PARs, consumer welfare rises modestly; without PARs, satellite share collapses and consumer welfare turns negative because exclusion bites.
- **MSG** illustrates a non-loophole cable-integrated RSN with strong efficiency gains and little foreclosure.
- **NESN** illustrates what happens when a non-integrated RSN is assigned to an integrated owner in the counterfactual: prices fall and carriage rises sharply with PARs, but gains are smaller and more fragile without PARs.

Economic message. The average treatment effect hides a lot of heterogeneity. Local footprint, WTP, and regulatory environment determine whether integration looks mostly efficient or mostly exclusionary.

TABLE IV
Simulated Market Outcomes for Selected RSNs[a]

		(i) No VI	(ii) VI PARs (vs. No VI)		(iii) VI No PARs (vs. No VI)	
		Level	$\%\Delta_{bl}$	$\%\Delta_{WTP}$	$\%\Delta_{bl}$	$\%\Delta_{WTP}$
Cable Integrated RSNs						
CSN PHIL	Cable Mkt Share	0.64	0.8%		1.8%	
Comcast		[0.62, 0.65]	[0.2%, 2.4%]		[0.6%, 4.0%]	
Pop 4.25M	Sat Mkt Share	0.18	−0.5%		−10.4%	
Footprint 90%		[0.17, 0.19]	[−3.3%, −0.2%]		[−14.8%, −0.5%]	
WTP $4.99	Cable Carriage	0.95	1.6%		0.4%	
		[0.62, 0.97]	[0.0%, 53.8%]		[−6.2%, 52.9%]	
	Cable Prices	54.31	−0.5%		0.9%	
		[53.28, 55.42]	[−1.5%, 0.9%]		[−1.4%, 1.8%]	
Foreclose: 85%	Aff Fees to Sat	2.26	3.6%		—	
		[1.00, 2.64]	[−9.4%, 7.0%]		—	
	Cable + RSN Surplus	30.19	0.2%	0.9%	1.1%	6.5%
		[14.57, 32.67]	[0.0%, 2.4%]	[0.3%, 13.7%]	[0.4%, 3.3%]	[3.0%, 20.5%]
	Satellite Surplus	4.29	−0.9%	−0.6%	−2.1%	−1.8%
		[1.26, 4.70]	[−3.4%, −0.4%]	[−2.4%, −0.5%]	[−4.8%, −1.1%]	[−4.5%, −0.9%]
	Consumer Welfare	31.21	0.6%	3.9%	−2.9%	−18.1%
		[16.82, 34.81]	[0.2%, 2.0%]	[1.4%, 12.7%]	[−3.3%, 1.5%]	[−21.8%, 9.9%]
	Total Welfare	65.69	0.3%	4.0%	−1.0%	−13.4%
		[31.14, 71.73]	[0.1%, 1.9%]	[2.0%, 25.2%]	[−1.1%, 1.1%]	[−15.6%, 14.7%]
MSG	Cable Mkt Share	0.63	3.3%		3.3%	
Cablevision		[0.62, 0.67]	[0.3%, 4.8%]		[0.2%, 4.7%]	
Pop 11.7M	Sat Mkt Share	0.18	−4.3%		−4.3%	
Footprint 42%		[0.17, 0.18]	[−7.1%, −0.4%]		[−8.1%, −0.4%]	
Pred WTP $2.32	Cable Carriage	0.68	10.5%		10.5%	
		[0.67, 0.87]	[−2.5%, 18.5%]		[−3.1%, 18.5%]	
	Cable Prices	59.40	−2.4%		−2.4%	
		[56.80, 60.81]	[−3.5%, 0.0%]		[−3.5%, 0.2%]	
Foreclose: 1%	Aff Fees to Sat	1.22	−3.3%		22.4%	
		[0.42, 1.28]	[−5.9%, 10.4%]		[17.1%, 55.4%]	
	Cable + RSN Surplus	30.64	0.3%	4.4%	0.5%	6.8%
		[14.61, 34.12]	[−0.1%, 0.6%]	[−1.6%, 7.4%]	[0.0%, 1.3%]	[0.4%, 14.6%]
	Satellite Surplus	4.16	−4.2%	−7.5%	−3.5%	−9.9%
		[1.24, 4.48]	[−7.2%, −0.5%]	[−12.1%, −0.9%]	[−8.5%, −1.2%]	[−14.3%, −2.4%]
	Consumer Welfare	33.80	2.1%	44.6%	3.0%	44.3%
		[18.38, 38.14]	[0.3%, 4.3%]	[4.4%, 66.3%]	[−0.4%, 4.3%]	[−6.3%, 66.0%]
	Total Welfare	68.60	1.6%	41.4%	1.4%	41.2%
		[32.06, 76.01]	[0.1%, 1.9%]	[3.4%, 60.9%]	[0.1%, 1.9%]	[2.5%, 60.7%]

(Continued)

TABLE IV—*Continued*

		(i) No VI	(ii) VI PARs (vs. No VI)		(iii) VI No PARs (vs. No VI)	
		Level	$\%\Delta_{bl}$	$\%\Delta_{WTP}$	$\%\Delta_{bl}$	$\%\Delta_{WTP}$
Non-Integrated RSN						
NESN	Cable Mkt Share	0.61	7.6%		9.4%	
*Comcast		[0.59, 0.65]	[1.6%, 11.2%]		[2.7%, 12.5%]	
Pop 5.20M	Sat Mkt Share	0.13	−7.8%		−22.3%	
Footprint 85%		[0.12, 0.14]	[−12.6%, −1.8%]		[−26.5%, −7.2%]	
WTP $6.91	Cable Carriage	0.92	6.2%		3.6%	
		[0.68, 0.98]	[0.0%, 33.1%]		[−0.5%, 38.1%]	
	Cable Prices	56.73	−4.7%		−3.9%	
		[54.24, 57.88]	[−6.6%, −0.5%]		[−6.0%, 0.6%]	
Foreclose: 96%	Aff Fees to Sat	3.32	3.1%		—	
		[1.23, 3.79]	[−12.6%, 16.9%]		—	
	Cable + RSN Surplus	28.38	0.9%	5.6%	2.0%	8.2%
		[15.68, 31.36]	[0.1%, 2.4%]	[0.9%, 16.6%]	[0.7%, 4.0%]	[5.4%, 16.7%]
	Satellite Surplus	2.96	−8.3%	−3.5%	−10.9%	−4.7%
		[0.84, 3.24]	[−13.2%, −1.8%]	[−5.5%, −1.3%]	[−13.9%, −3.0%]	[−6.3%, −1.7%]
	Consumer Welfare	28.36	6.4%	26.5%	3.5%	13.5%
		[15.54, 31.97]	[1.4%, 10.0%]	[8.2%, 40.8%]	[−1.7%, 7.1%]	[−9.0%, 29.2%]
	Total Welfare	59.70	3.1%	26.5%	2.0%	17.0%
		[29.79, 65.84]	[0.5%, 5.1%]	[7.8%, 43.7%]	[−0.2%, 4.5%]	[−2.5%, 37.5%]

[a]Scenarios (i)–(iii) correspond to the integration scenarios described at the beginning of Section 6. Beneath the RSN name is either the name of the RSN's owner (observed or, if non-integrated, assigned, which is denoted by *), the number of television households in the RSN's relevant markets, the MVPD owner's footprint (% of households passed) in the RSN's relevant markets, and the estimated mean consumer WTP for the RSN. Scenario (i) reports household weighted averages over all relevant markets for each RSN, where all levels except for market shares and cable carriage are in $/household/month. Scenarios (ii) and (iii) report changes from scenario (i), where $\%\Delta_{bl}$ (respectively, $\%\Delta_{WTP}$) represent changes from scenario (i) expressed as a percentage of changes in household weighted averages of levels (respectively, estimated mean consumer WTP for the channel). Affiliate fees to satellite are reported conditional on supply; missing values indicate foreclosure. 95% confidence intervals, constructed from 150 simulations, are reported below each figure; the fraction of simulations in which the RSN is predicted to foreclose at least one rival distributor under scenario (iii) is reported last under each RSN's name ("Foreclose: %"). Results for all other RSNs are contained in Appendix Tables A.V–A.VII.

5.7 Table V: average welfare effects across all RSNs

Read:

- With PARs, integration raises cable carriage by about **9.4%** and lowers cable prices by about **1.2%**.
- With PARs, consumer welfare rises by about **1.5%** and total welfare by about **0.7%**, equal to roughly **18.0%** and **16.8%** of an RSN's WTP, respectively.
- The total welfare gain with PARs averages about **$0.43 per household per month**.
- Without PARs, integrated RSNs foreclose in **4/26** cases and, conditional on supply, raise rival affiliate fees by about **18.0%** relative to the PAR benchmark.
- Relative to no integration, even integration without PARs still raises consumer welfare by about **1.3%** and total welfare by about **0.6%**, but satellite surplus falls by about **3.2%**.

TABLE V
Average Simulated Market Outcomes Across All RSNs[a]

	(i) No VI	(ii) VI PARs (vs. No VI)		(iii) VI No PARs (vs. No VI)		(iv) VI INRs (vs. No VI)	
	Level	$\%\Delta_{bl}$	$\%\Delta_{WTP}$	$\%\Delta_{bl}$	$\%\Delta_{WTP}$	$\%\Delta_{bl}$	$\%\Delta_{WTP}$
All RSNs							
Cable Mkt Share	0.60	2.1%		2.2%		0.1%	
	[0.59, 0.62]	[0.4%, 2.6%]		[0.6%, 2.6%]		[−0.2%, 0.9%]	
Sat Mkt Share	0.20	−2.8%		−8.4%		−0.6%	
	[0.20, 0.21]	[−2.8%, −0.4%]		[−4.7%, −2.8%]		[−2.6%, −0.0%]	
Cable Carriage	0.84	9.4%		8.0%		−1.4%	
	[0.68, 0.88]	[3.1%, 21.1%]		[0.0%, 18.1%]		[−4.4%, 0.9%]	
Cable Prices	55.18	−1.2%		−1.1%		0.1%	
	[54.25, 55.90]	[−1.5%, −0.1%]		[−1.4%, −0.8%]		[0.0%, 0.5%]	
Aff Fees to Rivals[b]	1.36	−6.7%		17.1%		18.9%	
	[0.34, 1.45]	[−12.2%, 4.4%]		[11.0%, 26.5%]		[12.1%, 28.8%]	
Cable + RSN Surplus	23.04	0.3%	3.1%	0.6%	5.9%	0.2%	1.9%
	[11.12, 24.17]	[0.1%, 0.6%]	[1.4%, 5.4%]	[0.2%, 0.9%]	[2.6%, 8.1%]	[0.1%, 0.5%]	[0.5%, 4.2%]
Satellite Surplus	5.18	−2.2%	−4.3%	−2.2%	−6.0%	−1.0%	−1.7%
	[1.56, 5.51]	[−3.1%, −0.5%]	[−6.7%, −0.3%]	[−3.9%, −1.0%]	[−8.6%, −1.1%]	[−1.5%, −0.6%]	[−2.5%, −0.6%]
Consumer Welfare	30.98	1.5%	18.0%	1.3%	16.2%	−0.2%	−1.7%
	[16.15, 34.47]	[0.3%, 1.8%]	[5.5%, 22.8%]	[−0.1%, 1.5%]	[−1.5%, 28.4%]	[−0.7%, −0.1%]	[−8.4%, 0.0%]
Total Welfare	59.13	0.7%	16.8%	0.6%	15.3%	−0.1%	−1.5%
	[27.59, 64.41]	[0.1%, 0.9%]	[5.4%, 22.2%]	[0.0%, 0.8%]	[0.8%, 18.7%]	[−0.2%, −0.0%]	[−6.4%, 0.1%]
# Foreclosed		4/26 [0, 9]					

[a]Average simulated market outcomes across the 26 RSNs in our analysis, weighted by the number of households in each RSN's relevant markets. Percentages are the averages of percentage changes across RSNs, weighted by the number of households in each RSN's relevant markets. "# Foreclosed" reports the number of RSNs that are not provided to rival distributors under (iii) VI No PARs. See Table IV and main text for additional details.

[b]"Aff Fees to Rivals" represents average affiliate fees (to the satellite MVPDs for cable-integrated RSNs, and to cable MVPDs and the rival satellite distributor for satellite-integrated RSNs) conditional on supply in each relevant scenario.

Economic message. The paper's headline is not that vertical integration is uniformly bad. It is that the efficiency gains are real, but access rules are what prevent those gains from being offset by exclusion and higher rival costs.

5.8 Table VI and Table VII: when foreclosure hurts the most

TABLE VI
WELFARE CHANGES FROM FORECLOSURE[a]

Percentage Change in Levels Between (ii) VI PARs and (iii) VI No PARs		
	%Δ Consumer Welfare	%Δ Total Welfare
Are Rival Distributors Excluded	−1.95***	−0.72***
	(0.53)	(0.22)
N	3,900	3,900
R^2	0.52	0.52

[a]Regression where the dependent variable is the percentage change in either consumer or total welfare in levels between integration scenarios (ii) and (iii) (with and without PARs in effect). Each observation is an RSN-counterfactual simulation (26×150). Specifications include RSN fixed effects. Standard errors clustered at RSN level, and *** represents significance at the 1% level.

TABLE VII
PROBABILITY OF EXCLUSION[a]

Exclusion of Rival Distributors by Integrated RSN (Without PARs)	
Footprint of Integrated Owner	0.67**
	(0.25)
WTP for RSN	0.07
	(0.05)
N	26
R^2	0.29

[a]Linear probability regression where the dependent variable is whether rival distributors are denied access to an RSN under integration scenario (iii) without PARs. Each observation is an RSN. Specification includes a fixed effect for whether the RSN owner is a cable operator. ** represents significance at the 5% level.

Read:

- Table VI shows that moving from integration with PARs to integration without PARs reduces consumer welfare by about **1.95%** and total welfare by about **0.72%** when rival distributors are excluded.
- Table VII shows that exclusion is more likely when the integrated owner's local footprint is high; the coefficient on footprint is 0.67 and statistically significant.
- The paper notes that the four foreclosing cable-owned RSNs all have footprints greater than about **85%**.

Economic message. Foreclosure is not random. It is most tempting when the integrated cable distributor already covers most of the households in the RSN's territory, so denying satellite access mainly steals viewers rather than sacrifices too many upstream affiliate fees.

6 Short summary

- The paper builds a structural model of multichannel television markets with household viewing, bundle choice, downstream pricing and carriage, and Nash-in-Nash affiliate-fee bargaining.
- The central empirical objects are RSN ownership, carriage patterns, ratings, bundle shares, affiliate fees, and the regulatory distinction created by PARs and the terrestrial loophole.
- Integration raises carriage of the owned RSN and lowers cable prices because integrated distributors partly internalize upstream profits.
- Internalization is substantial but incomplete: $\mu \approx 0.79$, so integration does not imply full joint-profit maximization.

- Observed loophole exclusion in Philadelphia and San Diego implies strong rival-foreclosure incentives.
- With PARs, integration is clearly welfare improving on average.
- Without PARs, some RSNs exclude rivals and others raise rival affiliate fees; average welfare still rises relative to non-integration, but rivals are harmed and the gains are much weaker where exclusion occurs.
- The policy lesson is that access rules matter greatly. They preserve the efficiency gains from integration while limiting the foreclosure margin.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that vertical integration in multichannel television markets has both pro-competitive and anti-competitive effects, and that the net effect is an empirical question rather than something that can be inferred from simple theory. In the RSN setting, the efficiency gains from lower double marginalization and greater carriage are quantitatively important. At the same time, the incentives to foreclose or overcharge rivals are also real, especially when program access rules are weak or absent.

7.2 Economic intuition

RSNs are high-value upstream inputs. Once a distributor owns one, lowering downstream prices and carrying the RSN more broadly can increase the joint profits of the integrated pair. That is the efficiency channel. But exactly because the RSN is valuable, denying it to a rival satellite distributor can also move subscribers toward the integrated cable platform. That is the foreclosure channel. The balance between these two forces depends on regulation, local footprint, and how much integrated firms actually internalize one another's profits.

7.3 Takeaway

The paper's biggest lesson is that *vertical integration is not the whole story; access regulation is.* When PARs are effective, integration mainly looks like an efficiency-enhancing coordination device. When PARs are absent, the same ownership structure can become an instrument for excluding rivals or raising their costs. That makes the paper a strong example of how structural IO can separate average efficiency gains from localized competitive harm.